

A typical network-centric application will have multiple end devices, gateways, switches and routers that are interconnected and generating data as well. When such a system is connected to the Internet, it adds further complications. System simulation tools are able to set up functional virtual platforms for heterogeneous processor architectures and allow devices to be configured on-the-fly.

"Fault injection capabilities can enable the system behaviour to be tested against vulnerabilities, and thereby address security concerns much ahead of time. To rig up such a complex system in a real world would not only have commercial implications but also technical challenges," adds Kiran Kumar S., field application engineer, Wind River.

The unique IoT design challenge

The Internet of Things (IoT) being a complex cyber physical system with umpteen number of nodes, the biggest design challenge is to ensure reliable and accurate interactions with the real world. Considering the scale at which IoT components would eventually be implemented in, designers have to not just look at whether individual systems work in tandem but also at how it is affecting the overall system when it is finally released into the real-world environment.

"Evolution of a simple design into a complex system is commonplace but we still struggle to manage complexity while accelerating innovation," according to NI Trendwatch 2014. It adds that this cyber-physical design challenge is amplified when designs span software, networks and physical processes. It also warns that failure to tackle this challenge can result in poor designs and stalled innovation. Using a model based design to design and validate systems will allow engineers to derive models from system specifications and environmental analyses.

What is new

Electronics simulation is not a new concept, but what is really exciting is the testing/debugging capabilities that these simulations bring in, which cannot be realised on real hardware. These add capabilities such as record and playback of an entire system simulation involving heterogeneous processor architectures and devices, fault injections to verify and validate system designs and exchange of simulation data for collaboration across team(s). Electronics simulation is also being used for system architecture analysis, much earlier than even hardware can be realised, thereby reducing iterative design cycles.

—Kiran Kumar S., field application engineer, Wind River

Semiconductor manufacturers like Atmel are using engineering simulation solutions from electronic design automation (EDA) vendors like ANSYS to model, analyse and optimise their IoT product portfolio. In a recent release, they stated that Atmel design teams around the world leveraged ANSYS HFSS, ANSYS RedHawk and ANSYS Totem to design and validate these complex systems on chips (SoCs) and platforms used across multiple IoT application segments. These simulation solutions helped enable Atmel to meet stringent power/performance requirements, ensure reliable operations across a wide range of frequencies and deliver products with tight time-to-market constraints.

The report adds, "These modelling techniques illuminate the interplay of practical design with formal models of systems that incorporate both physical dynamics and computation. Manual integration and deployment of these models are costly, time-consuming and error-prone. System design tools with the right levels of abstraction allow cyber and physical models to be automatically combined, simulated and deployed, and the same models are adaptable for requirement tracking and HIL verification."

Protecting your IP

Ever wanted to provide an early model of an amplifier to your client to ensure that he or she gets to test it in the system simulation and know if the design will win?

X-parameter generators are built to do exactly that. For instance,

Keysight's advanced design system (ADS) X-parameter generator allows you to create a model and request specification changes even before your hardware prototype is ready. It even lets you protect your IP by using Keysight's PNA non-linear vector network analyser to derive a very accurate model from measured hardware. Your customer can then use this virtual component in the simulation system to see just how well your design works out for him or her.

"Using tools like NI AWR visual system simulator you could either send that component you have designed as a blackbox, or use PXI systems to send signals through this, get the output and just send waveforms. This means that at the other end, you do not have to even see the waveforms. You just have to get these and input these into the system over there," explains Satish Mohanram, technical marketing manager, National Instruments.

Easier and better R&D

"We have this radio frequency (RF) tool chain that starts with AWR Microwave Office and visual system simulator, which can really help design microwave components and microwave circuitry," says Mohanram. This means you can design antennae and circuit boards that do all the complex signal processing and generate different waveforms. You could prototype the whole circuit and then certain parts of it can be virtually brought in as models. For instance, if you have built a PCB where all the transmission and reception is going on, you can modify one algorithm in